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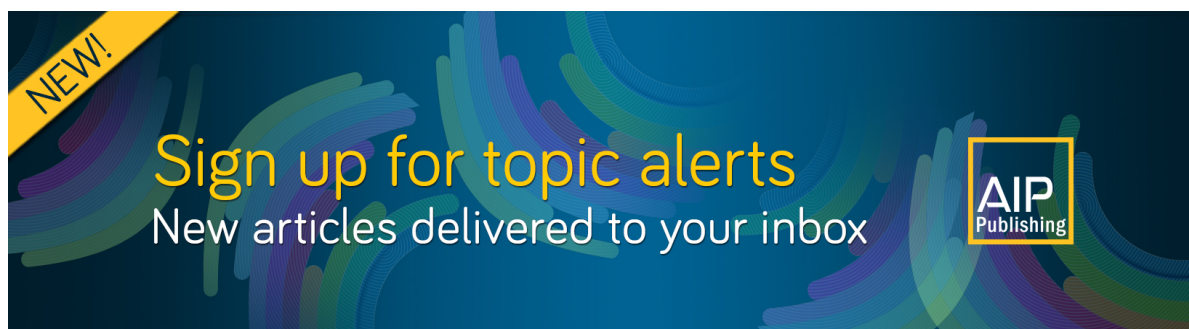
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ABSTRACT

Mineral dissolution in fractures is largely affected by the flow field within the fracture opening. The flow field maintains transport of the dissolved solutes and influences the concentration gradient and thermodynamic driving force for the mineral dissolution. In this study, we focus on investigating two factors that influence mineral dissolution in fractures: fracture wall roughness and interporosity fluid exchange or the flux through the fracture walls. The classical cubic law for the flow in parallel-plate channels cannot describe the flow in a fracture with rough permeable walls. The wavy, irregular shape of the rough boundaries can result in local flow features and might cause a shift of the overall flow and dissolution regime. At low Reynolds numbers, roughness simply increases the surface area available for mineral dissolution. Moreover, at higher Reynolds numbers, the inertial effects become important and the non-linear flow, flow instabilities, and reverse flow will form. With the creation of eddies and vorticities in the area adjacent to the walls, a flow dead zone forms that reduces the reactive surface area. In the presence of interporosity fluid exchange between the fracture and its surrounding rock, the non-zero fluid velocity on the fracture walls influences the local concentration gradient. In addition, the newly introduced fluid may facilitate or suppress the fracture wall mineral dissolution depending on its composition. This work highlights the compound effect of roughness and fluid flux through the walls on fracture mineral dissolution. For this purpose, the asymptotic solutions of the steady-state Navier–Stokes equations with non-zero velocity on the borders are used to determine the velocity field within the fracture opening. The quadratic and cubic corrections to Darcy's law are expected as a result of the wall roughness and the flow through the walls. The flow field is coupled with a transport module and a geochemical model (PHREEQC). As a test case, we investigate calcite dissolution in a single fracture at a different influx rate. For each flow rate, a flat fracture and two sets of rough profiles with and without the permeable walls were compared. The simulation results showed that compared to impermeable walls, the pervious walls result in a non-uniform non-periodic mineral dissolution along the fracture, which is more focused at the inlet. At low Reynolds numbers, the hotspots of dissolution are slightly shifted from the smaller cross sectional area to the larger cross sectional area. The effective reaction rate for mineral dissolution of fracture walls increases with an increase in fracture surface roughness. It is shown that at low Reynolds numbers, the permeable fracture walls can improve the effective reaction rate significantly. At relatively larger Reynolds numbers, the impact of flux through the walls on the effective reaction rate is less important, but for some roughness profiles, still not negligible.

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I. INTRODUCTION

Fractures are preferential pathways of fluid flow and solute transport in subsurface formations and, therefore, are important features that influence the fluid migration in applications such as geologic carbon sequestration, nuclear waste storage, geothermal energy extraction, reservoir acidizing, contaminant remediation, hydraulic

fracturing, and groundwater flow in karst systems. Fluid-rock chemical interactions (e.g., mineral dissolution/precipitation) especially in fractures can alter rock petrophysical properties and can, for example, impact the integrity of sealing formations or productivity of reservoir rocks.^{1–17} It should be noted that this impact is not necessarily adverse and in some cases, such as acidizing and geothermal energy extraction, is actually desirable.

Effective medium theory and multi-continuum models are widely used to model the flow in fractured media.^{18,19} These models treat the fractures as a high permeable porous zone and can account for fluid exchange or interporosity flow between the fracture and the matrix. The main drawback is that for open fractures, the application of Darcy's law oversimplifies the flow and may fail to capture transport limitation within the fracture toward its surface and on its surface.

Another common simplification in the study of fracture mineral dissolution involves describing the fluid flow in the fracture by cubic law,^{20–26} in which flux is linearly proportional to the pressure drop and the cube of the fracture aperture. The cubic law implies dominance of the viscous contribution and neglects the inertial effects. It is applicable for the fluid flow at low Reynolds numbers within fractures with no or small roughness and no fluid flux through the walls. Previous numerical, analytical, and experimental investigations that study the flow through a single fracture with rough surfaces^{27–39} have confirmed that the cubic law cannot accurately describe the non-linear characteristics of the flow in the fracture at high flow rates, when complicated flow phenomena such as formation of eddies and back flow may arise.^{31–36,38,39,77,79}

Various approaches have been proposed to improve flow modeling in single fractures such as applying the Forchheimer flow model, non-linear corrections to Darcy's law, or Darcy–Brinkman–Stokes formulation^{37,38,40–44,74–76,78} to address the impacts on the fluid flow of one or several of the factors, i.e., fracture geometry, flow rate (e.g., high Re number), and permeable walls. Rasoulzadeh and Panfilov⁴⁵ used a two-scale homogenization technique to capture the effect of the fracture wall corrugations and the flux through the walls on the flow field in the fracture. Obtaining asymptotic solutions to the complete set of Navier–Stokes equations, they concluded that the fluid exchange through the walls can create local flow instabilities and results in the formation of the reverse flow. They showed that in a rough fracture with pervious walls, the average pressure drop along the fracture includes quadratic and cubic corrections to the linear relation between pressure drop and velocity. The quadratic correction only exists in the case of pervious walls. The cubic correction corresponds to the wavy geometry. They also concluded that in the case of parallel-plate fractures, the cubic term is not significant compared to the linear and quadratic terms. Moreover, their results stated that the fracture's effective permeability decreases as the amplitude of the corrugations increases.

The complex flow characteristics that arise from rough geometry, high velocity, and permeable boundaries are expected to affect reactive transport in the fractures,^{21,22,41,46–56,73} as previous studies have shown that the geochemical alteration of the fracture surfaces depends on the interplay between chemical reactions at the mineral surfaces (e.g., interdependence of pH and dissolution rate) and the flow regime in the fracture.^{22,41,53–56}

Surface roughness increases the contact surface between the solid and the fluid and at the same time influences the flow field and the solute transport process via increasing the tortuosity and decreasing the permeability.^{33–35,38,39,51,52,57} Typically in geochemistry, the surface roughness factor (SRF), which is the ratio between the total surface area and the nominal surface area, is used to correct for the impact of irregular fracture walls on the reaction rate.^{58–60} Mineral accessibility to the reactive fluid due to transport limitations can be taken into account by introducing a scaling factor to

the SRF.^{15,61} However, the contribution of surface roughness to the reaction rate is not linearly proportional to the surface area, and the scaling factor also varies with velocity and roughness.⁵² This means that in some cases, the rough geometries need to be explicitly resolved in order to accurately predict the reaction rate.

In addition, most of the existing reactive transport models for mineral dissolution in fractures focus on the reactive processes at the fracture surface and do not account for mass transfer between the fracture and the porous matrix. Typically, advection in the main flow direction is coupled with transverse diffusion of the reactants in the near wall regions, i.e., the diffusive boundary layer.^{62–65} It should be noted that mass transfer through the walls can introduce a continuous fresh reactive fluid to the walls, which can accelerate mineral dissolution. Furthermore, as fluid enters or leaves the fracture, the local fluid velocity and concentration fields are altered, affecting subsequent reaction hotspots. Compared to an impermeable wall where the dominant transport mechanism is diffusion, the pervious walls allow advection to play a role in transporting chemicals near the fracture surfaces.

In this work, we aim at investigating the compound effects of fracture roughness and interporosity fluid exchange on mineral dissolution in a single fracture. This type of investigation is critical for improving our understanding of the role of micro-scale heterogeneity in the large-scale flow and dissolution processes and developing more representative effective models. While direct numerical simulation of the flow in fractures can be computationally expensive because of the small fracture aperture and high aspect ratio,^{66,67} we adopt the asymptotic analytical solution of Navier–Stokes equations for a parallel-plate wavy fracture with non-zero fluid velocity through the boundaries, obtained in Ref. 45. This steady-state flow field is used to compute the transport of chemicals in the fracture. Chemical reactions are simulated using PHREEQC.⁶⁸ PHREEQC is a geochemical solver that simulates a broad range of equilibrium reactions including ion exchange, surface complexation, gas dissolution, mineral reactions, etc. It also has a general kinetic formulation that allows modeling of non-equilibrium mineral dissolution and precipitation. The coupling between the asymptotic model of the flow and the reactive transport process provides a semi-analytical framework for single phase multi-component reactive transport in a single fracture with rough permeable walls.

The paper is organized as follows: Sec. II A details the mathematical model and analytical solutions for the flow in a rough fracture with pervious/impervious walls. Section II B presents the chemical reactions considered in this work, and Sec. II C documents the numerical solution of solute transport based on the flow field computed analytically. We present and discuss the simulation results in Sec. III and provide a conclusion on the study in Sec. IV.

II. METHOD AND MATHEMATICAL MODEL

The following modeling framework is set up to investigate the impact of rough permeable walls on mineral dissolution and, thus, fracture evolution.

A. Model of fluid flow in the fracture

Here, we summarize the key points of the analytical solution of the fluid flow in a rough fracture with permeable walls. A fracture

of length L and average opening width H is considered. The fracture roughness profile is given by $H\phi(X)$, where $\phi(X)$ is a dimensionless periodic function of wavelength l , the mean value $\frac{1}{2}$, and an average amplitude of a . X is the coordinate axis in the direction of the main longitudinal axis of the fracture and of the flow (Fig. 1). Y is the coordinate axis in the direction of the fracture depth which is assumed to be very large in this study. Z is the coordinate axis in the direction of the fracture opening width. In the case of $\phi(X) = \frac{1}{2}$, the wall is flat with no roughness. At the fracture inlet $X = 0$, we assume that $\phi(0) = \frac{1}{2}$.

The steady-state flow of an incompressible fluid with density ρ and dynamic viscosity μ within the fracture is considered. The dimensionless Navier–Stokes equations for the flow in the fracture are given as follows:

$$\omega \frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} = 0, \quad (1a)$$

$$\text{Re} \left(\omega u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \omega^2 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2}, \quad (1b)$$

$$\text{Re} \left(\omega u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial z} \right) = -\frac{1}{\omega} \frac{\partial p}{\partial z} + \omega^2 \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial z^2}. \quad (1c)$$

The inlet boundary condition is given by a constant influx. A constant uniform velocity field is imposed at the fracture boundary representing the fluid exchange through the porous walls. The flux through the interface of the fracture and the rock is assumed to be small compared to the main flux in the fracture.

The dimensionless boundary conditions for the upper half of the fracture aperture at the fracture's inlet and on the walls are given mathematically as follows:

$$\frac{1}{2} = \int_0^{\phi(0)} u(0, z) dz, \quad (2a)$$

$$v|_{z=\phi(x)} = -\omega v^*, \quad u|_{z=\phi(x)} = \omega u^*, \quad (2b)$$

$$\left. \frac{\partial u}{\partial z} \right|_{z=0} = 0, \quad v|_{z=0} = 0, \quad (2c)$$

$$\int_0^{\phi(x)} u(x, z) dz = F_0(x) + \omega F_1(x). \quad (2d)$$

In the above equations, the following dimensionless variables are used:

$$x = \frac{X}{L}, \quad z = \frac{Z}{H}, \quad p = \frac{P}{P^0}, \quad u = \frac{U}{U^0}, \quad v = \frac{V}{U^0}, \\ \omega u^* = \frac{U^*}{U^0}, \quad -\omega v^* = \frac{V^*}{U^0}, \quad \omega = \frac{H}{L}, \quad \varepsilon = \frac{\ell}{L}. \quad (3)$$

Here, x, z are dimensionless coordinates. The fracture dimension in the y direction is assumed to be very large. P, U , and V are the pressure, longitudinal fluid velocity, and transverse fluid velocity, respectively. p, u , and v are the corresponding dimensionless variables. U^0 is the mean velocity at the inlet section $x = 0$. The characteristic pressure $P^0 = \frac{\mu U^0 L}{H^2}$ is determined as the pressure drop along the length L caused by a Poiseuille flow at velocity U^0 . The dimensionless Reynolds number is defined as $\text{Re} = \frac{\rho U^0 H}{\mu}$. U^* and V^* are the uniform fluid velocity components on the walls. ωu^* and $-\omega v^*$ are the corresponding dimensionless variables that impose a constant influx condition on the fracture boundaries [Eq. (2a)]. ω is the aspect ratio (width to length) of the fracture [Eq. (3)]. Parameters ω and ε are small so that $0 < \omega, \varepsilon \ll 1$ and $\omega \sim \varepsilon^\alpha$, $\alpha \sim 1$. Note that $\omega \sim O(0.1)$ or lower and $u^*, v^* \sim O(1)$. The model framework allows the direction of the flux through the walls be varied by adjusting the values of u^* and v^* as long as they remain in unitary order of magnitude. For the purpose of the current study, we selected $u^* = v^* = 1$ for pervious fracture walls and $u^* = v^* = 0$ for impervious fracture walls. On the upper half of the fracture, this means that the vertical component of influx velocity is always toward the fracture main axis, and the horizontal component can enter or exit the fracture depending on the roughness slope. To acquire the analytical solution, the flux through the permeable walls is given as a fraction of the influx [Eq. (2b)], and the multiplying factor is ω . Equation (2c) describes the symmetry with respect to the fracture main axis. Equation (2d) captures local mass conservation, which shows that the flow at point x along the fracture length is the sum of the inlet flow and the fluid flow through the walls. $F_0(x), F_1(x)$ are given as

$$F_0(x) \equiv \frac{1}{2} + \int_0^x v^* d\tilde{x}, \quad F_1(x) \equiv u^*(\phi(x) - \phi(0)). \quad (4)$$

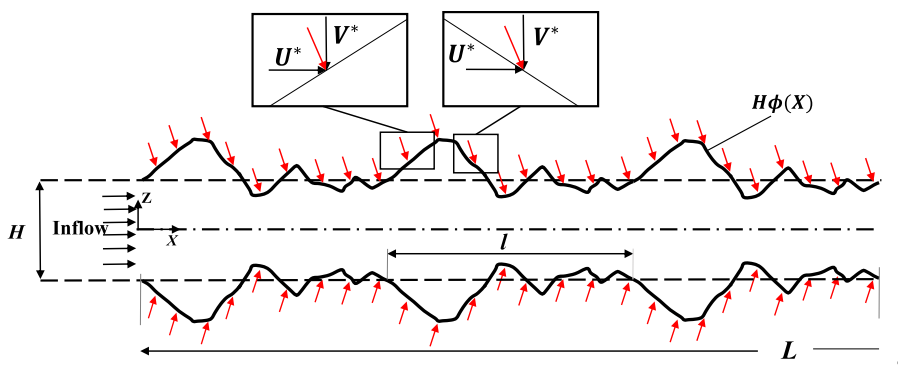


FIG. 1. Schematic of a fracture with self-similar rough walls and uniform fluid flux through the walls.

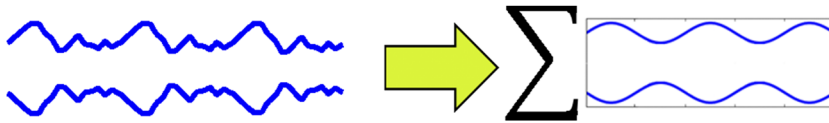


FIG. 2. Micro-structure of fracture surface roughness given by a piece-wise continuous function considered as over-laying of multiple sine waves.

The presented framework allows complex roughness geometries to be rigorously modeled. While the roughness function can be any piece-wise continuous function, the Fourier sine series of the roughness profile will be used in our computations to assure smooth first and second derivatives of $\phi(x)$ (Fig. 2),

$$\phi(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\varepsilon}\right), \quad b_n = \frac{1}{\varepsilon} \int_0^{\varepsilon} \phi(x) \sin\left(\frac{n\pi x}{\varepsilon}\right) dx. \quad (5)$$

The solution $\mathcal{U} = \{u, v, p\}$ is searched in the form of the asymptotic expansion with respect to small parameter ω ,

$$\mathcal{U} = \mathcal{U}_0(x, z) + \omega \mathcal{U}_1(x, z) + \omega^2 \mathcal{U}_2(x, z) + \dots \quad (6)$$

After substituting Eq. (6) in the problem [Eqs. (1) and (2)], we will obtain the following:

- For u_0 , v_0 , and p_0 as first terms in asymptotic power series given in Eq. (6),

$$\begin{aligned} \frac{\partial v_0}{\partial z} &= 0, \quad \frac{\partial p_0}{\partial z} = 0, \\ \text{Re } v_0 \frac{\partial u_0}{\partial z} &= -\frac{\partial p_0}{\partial x} + \frac{\partial^2 u_0}{\partial z^2}, \\ v_0|_{z=\phi(x)} &= 0, \quad u_0|_{z=\phi(x)} = 0, \\ \frac{\partial u_0}{\partial z} \Big|_{z=0} &= 0, \quad v_0|_{z=0} = 0, \\ \int_0^{\phi(x)} u_0 dz &= F_0(x). \end{aligned}$$

This directly gives $v_0 \equiv 0$.

- For u_1 , v_1 , and p_1 ,

$$\begin{aligned} \frac{\partial u_0}{\partial x} + \frac{\partial v_1}{\partial z} &= 0, \quad \frac{\partial p_1}{\partial z} = 0, \\ \text{Re} \left(u_0 \frac{\partial u_0}{\partial x} + v_1 \frac{\partial u_0}{\partial z} \right) &= -\frac{\partial p_1}{\partial x} + \frac{\partial^2 u_1}{\partial z^2}, \\ v_1|_{z=\phi(x)} &= -v^*, \quad u_1|_{z=\phi(x)} = u^*, \\ \frac{\partial u_1}{\partial z} \Big|_{z=0} &= 0, \quad v_1|_{z=0} = 0, \\ \int_0^{\phi(x)} u_1 dz &= F_1(x). \end{aligned}$$

- For u_2 , v_2 , and p_2 ,

$$\begin{aligned} \frac{\partial u_1}{\partial x} + \frac{\partial v_2}{\partial z} &= 0, \quad 0 = -\frac{\partial p_2}{\partial z} + \frac{\partial^2 v_1}{\partial z^2}, \\ \text{Re} \left(u_1 \frac{\partial u_0}{\partial x} + u_0 \frac{\partial u_1}{\partial x} + v_1 \frac{\partial u_1}{\partial z} + v_2 \frac{\partial u_0}{\partial z} \right) &= -\frac{\partial p_2}{\partial x} + \frac{\partial^2 u_0}{\partial x^2} + \frac{\partial^2 u_2}{\partial z^2}, \\ v_2|_{z=\phi(x)} &= 0, \quad u_2|_{z=\phi(x)} = 0, \\ \frac{\partial u_2}{\partial z} \Big|_{z=0} &= 0, \quad v_2|_{z=0} = 0, \\ \int_0^{\phi(x)} u_2 dz &= 0. \end{aligned}$$

We map the irregular geometry of the fracture walls to a flat geometry by introducing a new variable ζ , as shown in Fig. 3:

$$(x, z) \rightarrow (x, \zeta), \quad \zeta \equiv \frac{z}{\phi(x)}. \quad (7)$$

This mapping scheme is actually a change in the coordinate system which compresses or stretches the z direction such that the fracture border becomes straight. The points on the fracture wall correspond to $\zeta = 1$; and $\zeta < 1$ corresponds to the inner fracture points. All the spatial derivatives are rewritten with regard to this new coordinate system, given as

$$\frac{\partial}{\partial z} = \frac{1}{\phi} \frac{\partial}{\partial \zeta}, \quad \frac{\partial}{\partial x} = \frac{\partial}{\partial x} - \frac{\zeta \phi'}{\phi} \frac{\partial}{\partial \zeta}, \quad (8)$$

where $\phi' = \frac{d\phi}{dx}$. An advantage of this mapping strategy is to avoid challenges related to unstructured gridding because of the fracture non-regular wall geometry. In the mapped domain, as it is shown in Fig. 3, a structured grid is used for the numerical model, and a simple numerical scheme, i.e., finite difference can be used for the transport equation. All the scalar quantities such as concentration $C(x, \zeta)$ can be mapped back to the (x, z) system only by change in ζ to the corresponding z . The limit $\omega \rightarrow 0$ corresponds to the elimination

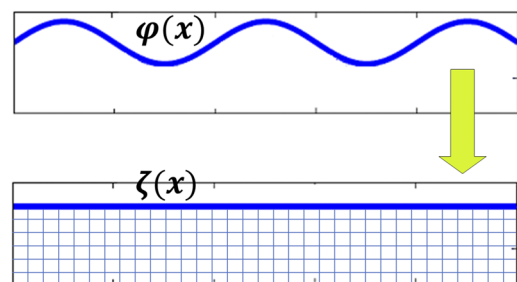


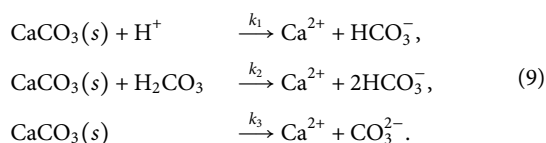
FIG. 3. Non-regular geometry of walls mapped to a flat geometry and the corresponding structured grid.

of the transverse variable ζ and, thus, the homogenization along z . Thus, the first term in the asymptotic series expansion in Eq. (6) is expected to be equal to the averaged value over the cross section of the fracture.

At the limit $\varepsilon \rightarrow 0$, the homogenized behavior of the system along x will be obtained.⁴⁵ The average behavior of this system along x is not of interest in this work as we investigate the impact of fracture roughness on mineral dissolution at the pore scale. The complete form of the asymptotic solution for the flow field including u , v , and p is given in Appendix. More detailed derivation and analyses of the flow in the parallel-plate fracture and an axisymmetric fracture can be found in Ref. 45.

B. Geochemical modeling

Dissolution of a single mineral, calcite, is investigated in this study as a test problem. Nevertheless, the modeling framework presented in this paper can include different mineral reaction scenarios. Calcite is a fast-reacting mineral that is abundant in subsurface formations. The carbonate-rich (calcite content about 50%) solid phase is completely contained within the porous matrix. A preferential flow takes place in the fracture. Initially, the domain is saturated with brine of pH = 6.83 and low concentration of aqueous calcium ion $\text{Ca}^{2+}(\text{aq})$. A 0.05M NaCl brine acidified by 0.01M CO_2 (i.e., $p\text{CO}_2 = 0.3$ bar) under ambient pressure and temperature conditions, which has a pH of 4.2 and is unsaturated with respect to calcite, is injected from the left through the fracture inlet and is transported out of the domain across the fracture outlet on the right. The mineral dissolution takes place at the interface with the porous host matrix at $z = \phi(x)$. We consider the case where all aqueous reactions reach equilibrium instantly. Calcite dissolution is treated kinetically and described by the following three parallel reactions, where the first elementary reaction shows the dependence of reaction rate on pH and the second elementary reaction depends on the dissolved CO_2 concentration,^{69,70}



All reactions (aqueous equilibrium reactions and kinetic mineral reactions) were simulated using PHREEQC, an open-source general-purpose geochemical simulator developed by the US Geological Survey.⁶⁸ Using PHREEQC as the chemical reaction solver provides flexibility in terms of accounting for complex reaction networks and realistic geochemical reaction rates, while in this study, we consider a relatively simple geochemical system; the model framework would allow simulations of a wide variety of chemical reactions. Constitutive relations for reaction rate laws and mass laws are defined through the PHREEQC input file.

The simulations are run until steady state has been reached. Two parameters are calculated to investigate the local and overall mineral dissolution. First, the point-wise cumulative reaction over time, which can illustrate the expected cumulative mineral dissolved at each point on the wall and the local reaction hotspots on the rough fracture surface. It should be noted that given the initial

transient period is short compared to the overall simulation time, this parameter is reflective of the steady-state local reaction rate. Second, the average calcium (Ca^{2+}) concentration at the fracture outlet, the same as the one introduced in Ref. 52, to quantify the bulk effective reaction rate is

$$R_{\text{bulk}} = \int_0^{H\phi(L)} c_i(L, z) u(L, z) dz, \quad i = \text{Ca}^{2+}. \quad (10)$$

C. Transport modeling

In the simulations performed here, we assumed that the flux through the wall has the same fluid chemistry as that of the inflow at the inlet. This represents the scenario when there is a background flow in the matrix and the upstream is reactive. The modeling framework can be used to simulate the scenarios when the influx through the inlet and that through the fracture walls have different fluid chemistry. The transient transport of a single phase reactive multi-component system is described by the advection–diffusion equation in the dimensionless form as

$$\frac{\partial c_i}{\partial \tau} + \text{Pe} \left(\omega u \frac{\partial c_i}{\partial x} + v \frac{\partial c_i}{\partial z} \right) = \omega^2 \frac{\partial^2 c_i}{\partial x^2} + \frac{\partial^2 c_i}{\partial z^2}, \quad i = \text{Ca}^{2+}, \text{H}^+, \text{HCO}_3^-, \text{CO}_3^{2-}. \quad (11)$$

This system of equations in (x, ζ) rewrites to

$$\begin{aligned}\frac{\partial c_i}{\partial \tau} + \text{Pe} \left(\omega u \left(\frac{\partial c_i}{\partial x} - \frac{\zeta \phi'}{\phi} \frac{\partial c_i}{\partial \zeta} \right) + \frac{v}{\phi} \frac{\partial c_i}{\partial \zeta} \right) \\ = \omega^2 \left(\frac{\partial}{\partial x} - \frac{\zeta \phi'}{\phi} \frac{\partial}{\partial \zeta} \right)^2 c_i + \frac{1}{\phi^2} \frac{\partial^2 c_i}{\partial \zeta^2}.\end{aligned}\quad (12)$$

Here, c_i is the concentration of component i , $\text{Pe} = \frac{U^\circ H}{D_m}$ is the dimensionless Peclet number, and D_m is the dimensional molecular diffusion coefficient in water. τ is the dimensionless time with respect to the characteristic time for Ca^{2+} diffusion through the fracture width $\tau = \frac{tD}{H^2}$, and u and v are fluid velocities computed from the steady-state analytical solution for the flow field given in Sec. II A. The advection–diffusion equation is solved numerically on the mapped coordinate system (x, ζ) using a finite difference scheme of the second-order central difference approximation in space and a first-order forward Euler scheme in time. The time step is constrained by the Courant–Friedrichs–Lewy criterion. A schematic of the spatial structured grid is shown in Fig. 3.

D. Coupled reactive transport model

An Application Programming Interface (API) is developed for couple flow, transport, and chemical reactions. A Sequential Non-Iterative Approach (SNIA) scheme is employed for coupling the transport step and the chemical reactions (Fig. 4). The sequential non-iterative approach (SNIA) is a commonly used coupling scheme

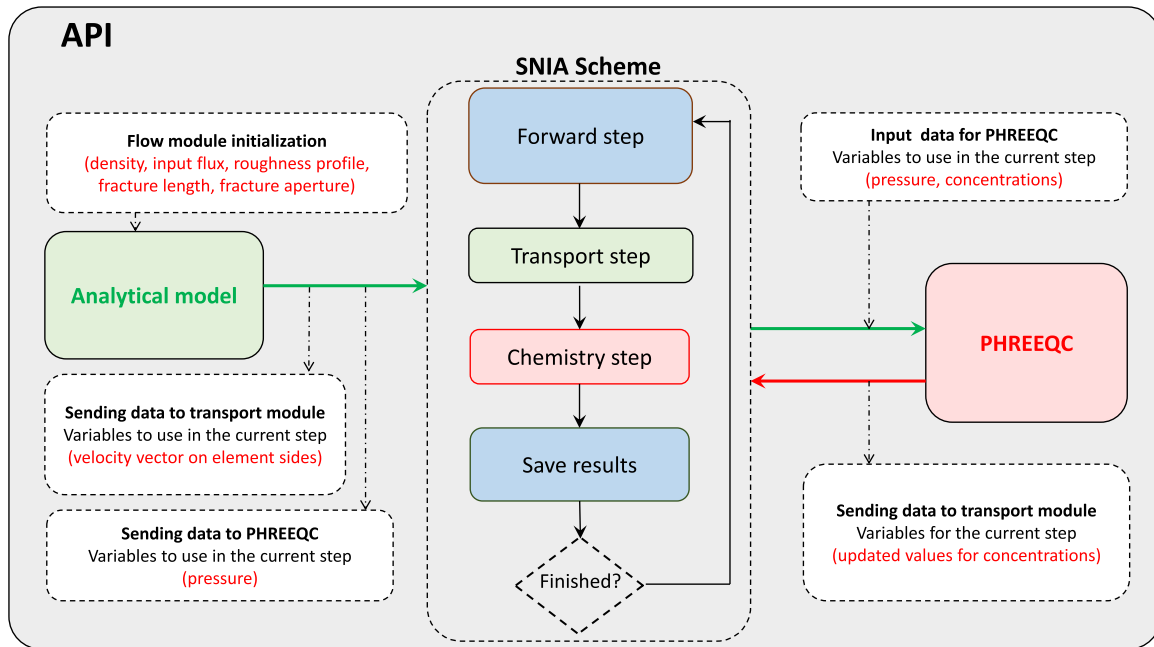


FIG. 4. Schematic of the Application Programming Interface (API) and the Sequential Non-Iterative Approach (SNIA) coupling scheme used for coupling flow, mineral dissolution, and solute transport.

in reactive transport modeling. Carrayrou *et al.*⁷¹ compared five coupling schemes for a reactive transport benchmark. In their work, the SNIA proved good efficiency [less central processing unit (CPU) time] and reasonable error magnitude.

First, the flow model is initialized for the given fluid properties, fracture roughness profile, and the influx rate. The flow field is solved analytically to compute the velocity at the midpoint of the element edges. The pressure drop along the fracture is also calculated. The velocity vector is used in the transport module to simulate solute transport (Fig. 4). The transport module is initialized based on the initial distribution of the mineral species. For each step of transport, the advective terms and diffusive terms are calculated to compute the composition of the species at each cell. The kinetic reactions and equilibrium reactions are computed in the reaction module. The basic function of the reaction module, PHREEQC, is to update the component concentrations at the center of the elements on the wall computed from the transport module and return the updated value to the transport module. The simulations presented in this paper do not consider evolution of the fracture–matrix interface during the dissolution process because we focus on the steady-state reaction rate and also partly because the matrix has 50% of non-reactive minerals remaining.

III. RESULTS AND DISCUSSION

Mineral dissolution in a fracture with flat walls is compared to mineral dissolution in fractures with two sets of roughness profiles: a single sine wave and a self-similar geometry by overlaying five sine waves, which is as follows:

$$\begin{aligned}\phi_{\text{flat}}(x) &= \frac{1}{2}, \\ \phi_{\text{single sine}}(x) &= \frac{1}{2} + 0.15 \sin\left(\frac{2\pi x}{\varepsilon} + \frac{\pi}{2}\right), \\ \phi_{\text{multiple sine}}(x) &= b_1 \sin\left(\frac{2\pi x}{\varepsilon} + \frac{\pi}{2}\right) + b_3 \sin\left(\frac{3\pi x}{\varepsilon} + \frac{\pi}{2}\right) \\ &\quad + b_4 \sin\left(\frac{4\pi x}{\varepsilon} + \frac{\pi}{2}\right) + b_6 \sin\left(\frac{6\pi x}{\varepsilon} + \frac{\pi}{2}\right) \\ &\quad + b_9 \sin\left(\frac{9\pi x}{\varepsilon} + \frac{\pi}{2}\right),\end{aligned}\quad (13)$$

with

$$\begin{aligned}b_1 &= -0.0955, \quad b_3 = 0.0608, \quad b_4 = -0.0477, \\ b_6 &= -0.0318, \quad b_9 = 0.0211.\end{aligned}$$

We selected $\varepsilon = 0.03$ as the length of one period of the roughness geometry. The surface roughness factor is defined as the ratio of the arc length of the roughness to the fracture length L . The periodic roughness implies

$$\text{SRF} = \frac{1}{\varepsilon} \int_0^\varepsilon \sqrt{1 + (\phi'(x))^2} dx. \quad (14)$$

For the three roughness profiles investigated here, the surface roughness factor is

$$\begin{aligned}\text{SRF}_{\text{flat}} &= 1, \\ \text{SRF}_{\text{single sine}} &= 1.1975, \\ \text{SRF}_{\text{multiple sine}} &= 1.098.\end{aligned}\quad (15)$$

For each fracture, impervious (a no slip condition on the fracture surface $u^* = v^* = 0$) and pervious walls ($u^* = v^* = 1$) are investigated. The impervious wall assumption is common in numerical studies of fracture mineral dissolution and is likely to be valid in tight formations of an ultralow permeable matrix. The flow field in the fracture is computed at Reynolds numbers between $Re = 0.1$ and $Re = 55.0$, within the range of the reported values in the referred studies.^{52,72} The high Reynolds number $Re = 55.0$ considered here is the lowest value for which the reverse flow forms in the single sine wave rough fracture with the fracture aspect ratio of this study. To investigate the impact of fluid velocity, fracture surface geometry, and interporosity fluid exchange on the dissolution process, we compare the streamlines Ψ , dimensionless velocities u and v , aqueous calcium ion Ca^{2+} concentration, local cumulative calcite dissolution, and effective reaction rate for different scenarios simulated. Cumulative calcite dissolution presented is normalized to the maximum value in the case of solid (impervious) walls of the same geometry and at the same Reynolds number. All plots are presented for an upper half of the fracture geometry given that the fracture is assumed to be symmetric with respect to the x axis.

A. Flow field and dissolution

1. Flat fracture with no roughness

At three relatively low Reynolds numbers $Re = 0.1$, 1.0 , and 10.0 , similar streamline (Ψ) structures were observed, and thus, only one set of the flow field results corresponding to $Re = 0.1$ is presented here (Fig. 5). In the horizontal parallel-plate fracture with solid impervious walls [Fig. 5(a)], the streamlines Ψ are horizontal and follow the fracture wall geometry due to significant viscous effects compared to inertial effects. The longitudinal velocity u is maximum at the center of the fracture (i.e., axis of symmetry) and is zero at fracture walls following the no slip condition. The transverse velocity v is zero everywhere. In contrast, when fluid exchange through the fracture walls is permitted, the streamlines Ψ are no longer parallel to the fracture walls due to the inertial effects near the walls. The longitudinal velocity, u , is non-zero near the walls, but small and is maximum at the fracture centerline. The longitudinal velocity increases along the flow direction in accordance to

mass conservation. The transverse velocity v is directed toward the fracture axis of symmetry and is zero on the axis.

As the Reynolds number increases to a relatively high value, $Re = 55.0$, the inertial term becomes increasingly important. However, for the flat geometry, this increase in the inertial effect does not result in an observable change in the flow patterns and, therefore, is not plotted. Previous studies showed that a much higher flow velocity (e.g., $Re \gg 500.0$) is needed for the reverse flow to be observed.⁴⁵

The distribution of calcium Ca^{2+} concentration in the fracture opening with impervious/pervious flat walls at various Reynolds numbers, $Re = 0.1$, 1.0 , 10.0 , and 55.0 , is shown in Fig. 6. In the case of solid walls, the calcium concentration peaks at the solid interface and is almost zero at the central region of the fracture [Fig. 6(a)]. At the impervious fracture surface, the sole transport mechanism is diffusion, resulting in a narrow solute transport boundary layer. The width of the boundary layer depends on the Reynolds number. At the low Reynolds number $Re = 0.1$, the diffusive terms are comparable to advective terms and, thus, are significant in a wider region [Fig. 6(a)]. As the Reynolds number increases, the width of the transport boundary layer decreases as a result of domination of advective terms over diffusive terms [e.g., Figs. 6(c), 6(e), and 6(g)].

In the case of pervious walls, a non-zero advective flow transports the calcium Ca^{2+} released at the fracture surface into the fracture. The combined advective and diffusive transport widens the boundary layer [Fig. 6(b)] defined visually by the large concentration contrast. The width of the boundary layer is less affected by the increase in the Reynolds number than the impervious case, indicating that the advective contribution is primarily from the influx through the walls [Figs. 6(d) and 6(f)]. At $Re = 55.0$, the narrow transport boundary layer is a result of large longitudinal velocity, and it grows because of the increasing vertical advective transport from the influx through the permeable walls [Fig. 6(h)]. Figure 7 highlights the impact of the flux through the walls on the cumulative calcite dissolution along the flat fracture. This is the cumulative amount of calcite expected to be removed from the solid rock at every grid point from beginning the simulation until the calcium concentration at the fracture outlet reaches the steady state. At all Reynolds numbers, the cumulative dissolved calcium along the fracture with pervious walls is slightly larger than that in the case of

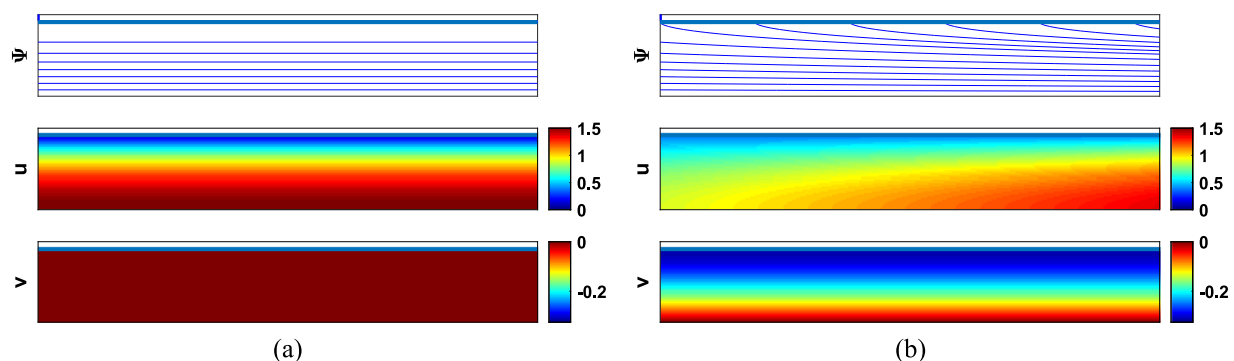


FIG. 5. Flow field within half of a fracture with a flat wall geometry at a low Reynolds number, $Re = 0.1$: impervious walls (a) and pervious walls (b). Similar structure of the flow field applies to $Re = 55.0$ with a larger magnitude of longitudinal velocity u . The thick solid blue line represents the upper fracture surface border.

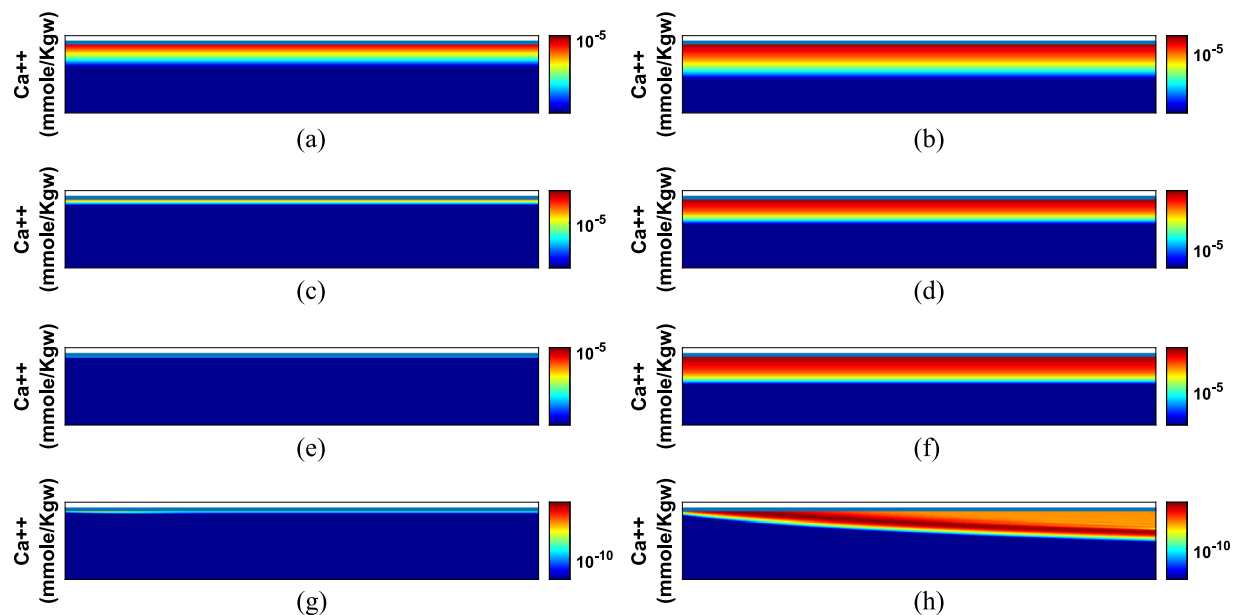


FIG. 6. Calcium ion concentration within half of a fracture with a flat wall geometry at different fluid velocities, $Re = 0.1$: impervious (a)/pervious (b) walls, $Re = 1.0$: impervious (c)/pervious (d) walls, $Re = 10.0$: impervious (e)/pervious (f) walls, $Re = 55.0$: impervious (g)/pervious (h) walls. The thick solid blue line represents the upper fracture surface border.

impervious walls. The point-wise cumulative dissolution in the fracture with pervious walls at $Re = 0.1$ is overall 1.2% higher at every grid point than that of the fracture with impervious walls [Fig. 7(a)]. At $Re = 1.0$, the dissolution improvement is $\sim 2.0\%$. However, this increasing trend does not continue for higher Reynolds numbers. At $Re = 10.0$, the cumulative mineral dissolution in the case of pervious walls is only 0.3% higher than that of the impervious wall simulation [Fig. 7(c)]. This difference can be considered as negligible as it is comparable to numerical errors from the finite difference scheme or the non-iterative sequential scheme evident close to the inlet. The comparable dissolution is also observed at the highest Reynolds number simulated [$Re = 55.0$, Fig. 7(d)]. At low Reynolds numbers, $Re = 0.1$, 1.0 , and 10.0 , when the walls are impervious to the fluid, the cumulative point-wise dissolution curves [Figs. 7(a)–7(c)] show a uniform pattern along the fracture, while in the case of porous walls, the dissolution is higher close to the fracture inlet and progressively becomes uniform along the fracture [Figs. 7(a)–7(c)]. At $Re = 55.0$, the progressive decrease of cumulative mineral dissolution along the flat fracture is more visible [Fig. 7(d)]. As it is shown in this figure, the dissolution is focused at the fracture inlet.

2. Fracture with single sine wave roughness

To investigate the combined impact of the wall roughness and the flux through the fracture walls, roughness is added to the set of simulations reported above. A single sine wave with the mean value $\frac{1}{2}$ is considered. Note that both the analytical solutions for flow, Eqs. (A1)–(A3), and the numerical solution of the transport equation, Eq. (12), were derived for the mapped coordinate system with

structured grids. The results are then converted back to the original coordinate systems for plotting.

The results for streamlines, longitudinal velocity, and transverse velocity are similar for $Re = 0.1$, $Re = 1.0$, and $Re = 10.0$, and thus, we only show the results for $Re = 0.1$ (Fig. 8). At these Reynolds numbers, when the walls are impervious to the fluid, the streamlines Ψ follow the wall geometry as the viscous effect is dominant [Fig. 8(a)]. The longitudinal velocity u along the centerline of the fracture is higher where the cross sectional area is smaller. Transverse velocity v maintains the required velocity component to follow the fracture non-regular geometry and may be upward or downward depending on the spatial location. Both velocities are zero on the fracture surface. With porous walls [Fig. 8(b)], the streamlines Ψ deviate slightly from the fracture geometry and the longitudinal velocity u increases progressively along the fracture. The longitudinal velocity is maximum where the fracture aperture is small. The transverse velocity increases along the fracture but stays much smaller than the longitudinal velocity. The absolute value of the transverse velocity is asymmetric with respect to the vertical axis of the symmetry, i.e., the vertical line crossing the peak of the sine wave, because of the influx. Both velocities are non-zero on the fracture surface.

At $Re = 55.0$, the increased contribution from the inertial term is evident in the streamlines, showing the formation of reverse flow or vorticities (Fig. 8). For fractures with impervious walls, the vorticities create a so-called dead zone to the fluid flow near the fracture surface [Fig. 8(c)]. With pervious walls, the inertial effects create a reverse flow in the near wall area [Fig. 8(d)]. The fluid that enters the fracture through the interface flows backward with respect to the general flow direction, then flows toward the fracture outlet.

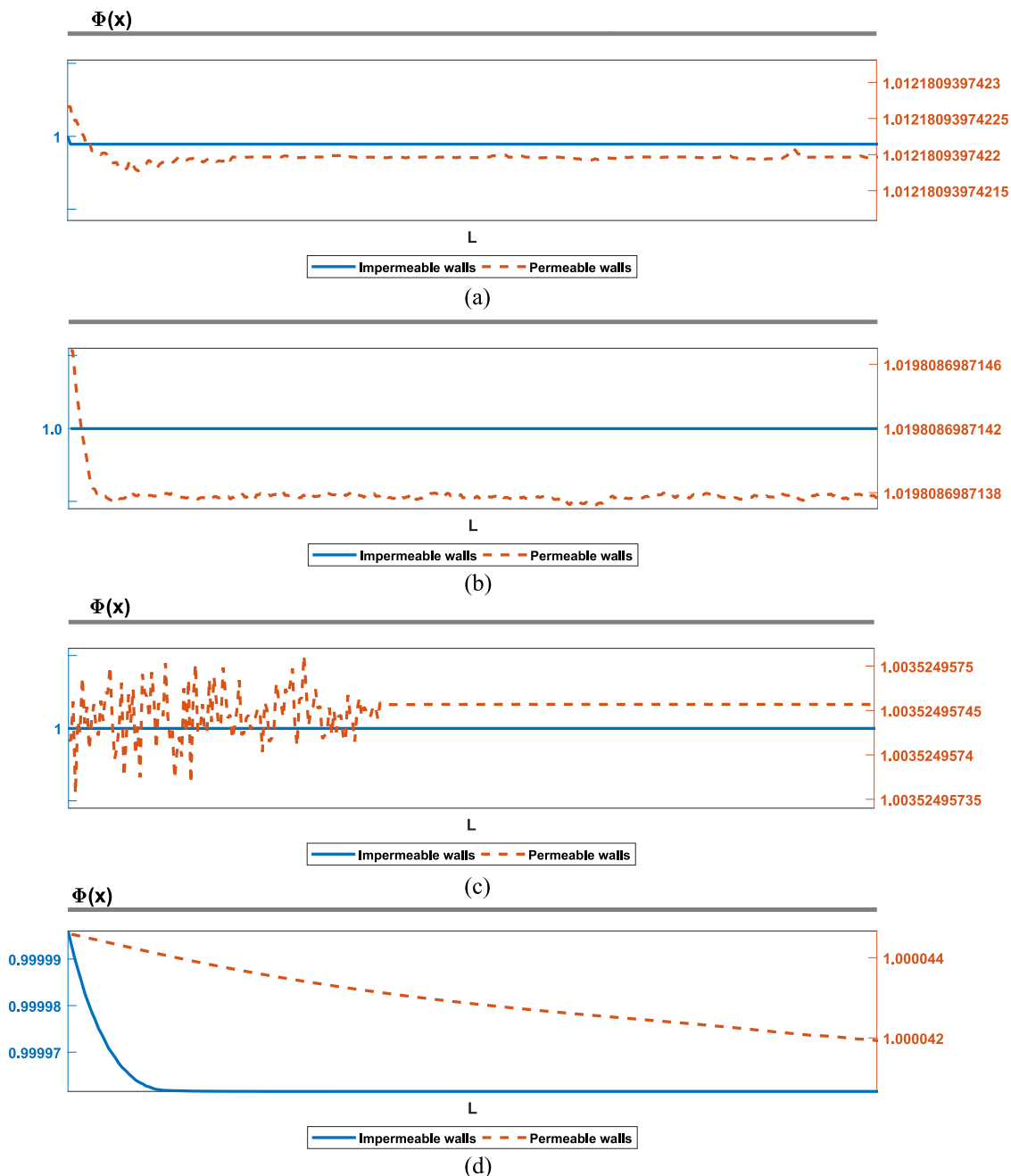


FIG. 7. Cumulative calcite dissolution on the flat impervious and pervious fracture walls normalized with respect to the maximum of the corresponding impervious fracture wall simulation, $Re = 0.1$ (a), $Re = 1.0$ (b), $Re = 10.0$ (c), and $Re = 55.0$ (d). The blue solid line reads from the left axis and the dashed orange line reads from the right axis.

At low Reynolds numbers, $Re = 0.1$, 1.0 , and 10.0 , the transport of calcium takes place within a narrow boundary layer near the fracture surface (Fig. 9). The boundary layer, or transport boundary layer, is where the calcium concentration is significant. The width of this boundary layer remains almost uniform along the fracture for both impervious and pervious walls, for $Re = 0.1$ [Figs. 9(a)

and 9(b)], for $Re = 1.0$ [Figs. 9(c) and 9(d)], and for $Re = 10.0$ [Figs. 9(e) and 9(f)]. With impervious walls and at $Re = 0.1$, the diffusive term of transport is more important and the solute transport boundary layer is mainly formed by the diffusion [Fig. 9(a)]. With permeable walls, the transport boundary layer is wider [Fig. 9(b)]. As the Reynolds number increases to $Re = 1.0$ and 10.0 , the

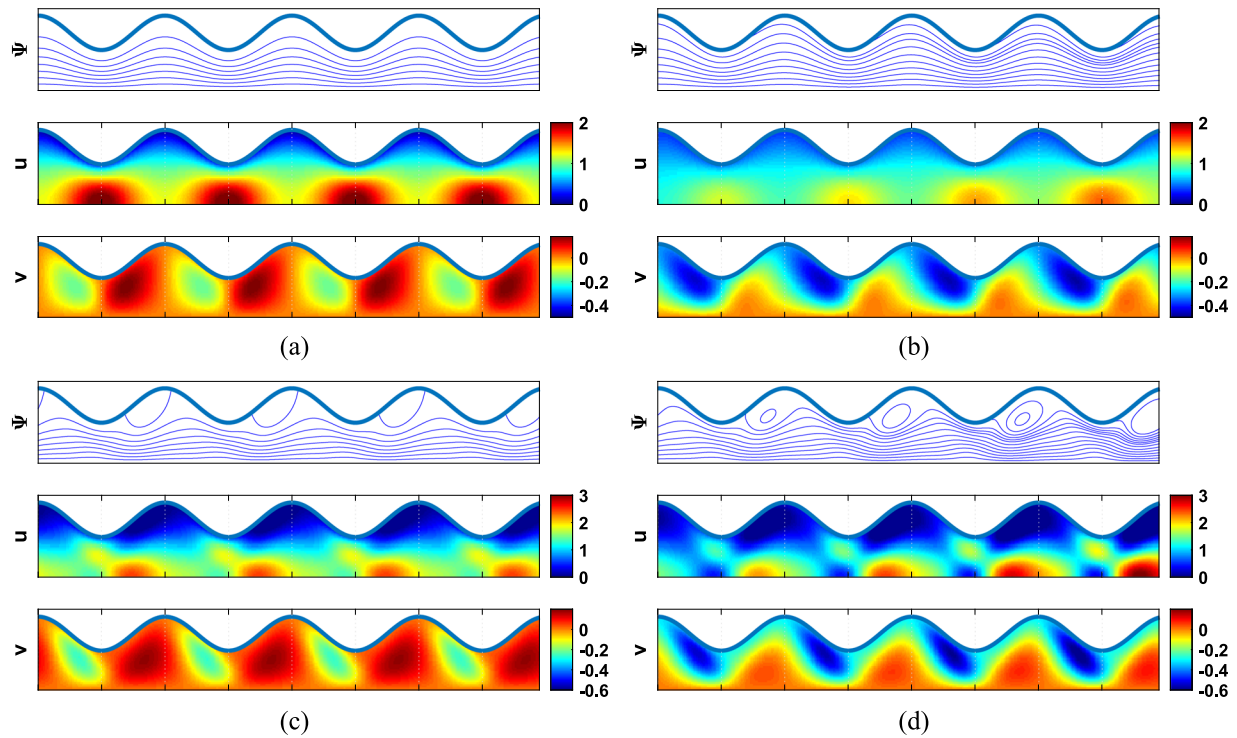


FIG. 8. Flow field within half of a fracture with a single sine wave wall geometry at low Reynolds numbers, $Re = 0.1$: impervious (a)/pervious (b) walls, and high Reynolds number, $Re = 55.0$: impervious (c)/pervious (d) walls. A higher density for plotting the contour lines in figures [(b) and (d)] is selected to show the streamlines starting from the fracture surface. The thick solid blue line represents the upper fracture surface border.

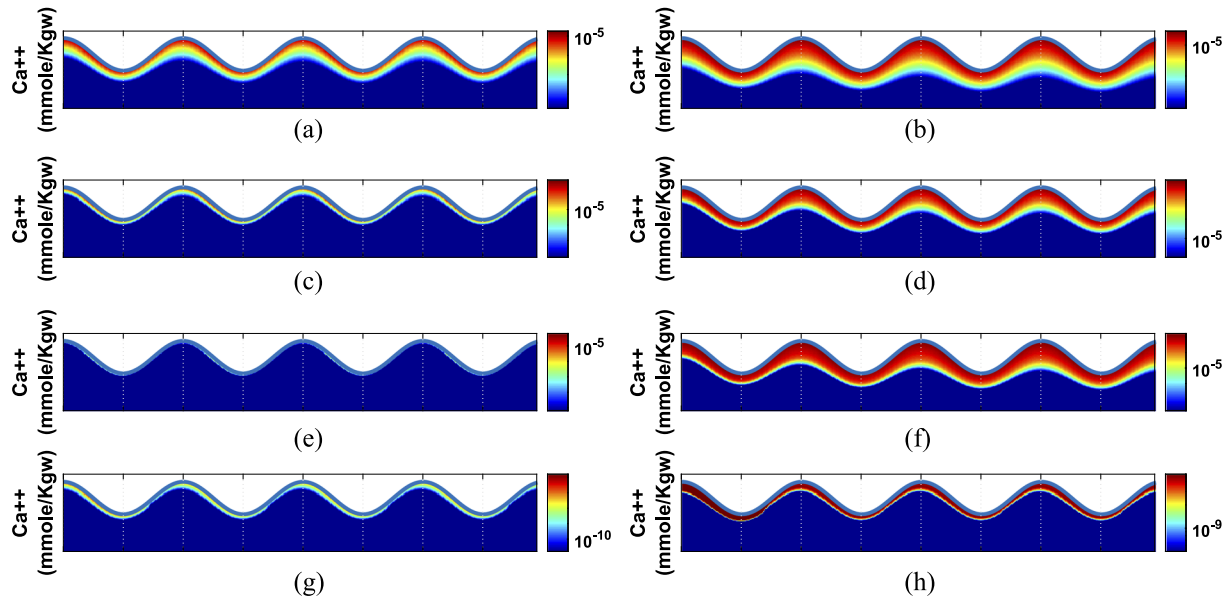


FIG. 9. Calcium ion concentration within half of a fracture with a sine wave roughness morphology geometry at different fluid velocities, $Re = 0.1$: impervious (a)/pervious (b) walls, $Re = 1.0$: impervious (c)/pervious (d) walls, $Re = 10.0$: impervious (e)/pervious (f) walls, $Re = 55.0$: impervious (g)/pervious (h) walls. The thick solid blue line represents the upper fracture surface border.

width of the boundary layer is reduced when walls are impervious [Figs. 9(a), 9(c), and 9(e)], while the boundary layer width is merely affected in the case of pervious walls [Figs. 9(b), 9(d), and 9(f)].

At a higher Reynolds number, $Re = 55.0$, in the fracture with impervious walls, the large longitudinal velocity limits the width of the solute transport boundary layer, and the vorticities limit the transport of the solute further because within the vorticity zone, the fresh solute cannot replace the existing fluid in place through advection [Fig. 9(g)]. Moreover, the solute transport boundary layer in zones adjacent to vorticities is narrower [this is hardly visible in the graph for Ca^{2+} in Fig. 9(g)]. When the surface of a rough fracture is pervious to the fluid, the small non-zero transverse and longitudinal velocity in the reverse flow zone area creates larger gradients of solute concentration, and as a result, the solute Ca^{2+} transport boundary layer gets wider [Fig. 9(h)], compared to the impervious case [Fig. 9(g)].

Figure 10 compares the cumulative mineral dissolution for the case of impervious and pervious walls in the single sine wave geometry at different Reynolds numbers. On the top of this graph, the roughness profile, $\Phi(x)$, is plotted in gray to assist relating the reaction hotspots to the geometry of the fracture. Overall, when fracture walls are impervious, the cumulative point-wise dissolution curve has a similar shape to the roughness profile (with various degrees of shifts), while in the case of porous walls, it shows a decreasing trend along the fracture.

In the case of solid walls and at a low Reynolds number $Re = 0.1$, the reaction hotspots are at locations where the cross sectional area is smaller and the longitudinal velocity is larger. This is consistent with the saturation index profile reported in Ref. 52. In their work, a higher saturation index was reported at the domes of rough fractures, which means lower dissolution at wider fracture cross sections and more dissolution at narrow fracture cross sections. Based on these observations, one might expect that over time, the roughness will reduce, and as a consequence, there will be less resistance to the fluid to pass and an increase in permeability. At $Re = 0.1$, mineral dissolution in the case of pervious walls is 1.2% larger than that in the impervious case [Fig. 10(a)]. At $Re = 1.0$, the cumulative mineral dissolution in the case of pervious walls is approximately 2.0% higher than that in the case of impervious walls [Fig. 10(b)]. For the impervious case, the reaction hotspots (the local maximum cumulative dissolution along the fracture wall) does not correspond anymore to the minimum cross sectional area, as was the case at a lower Reynolds number, $Re = 0.1$, and is displaced backward with respect to the flow direction. This shift in the reaction hotspots is seen both for impervious and pervious walls, for the single sine wave profile [Fig. 10(b)]. In this case, one might expect an increase in fracture wall roughness as a result of dissolution. As the Reynolds number increases further ($Re = 10.0$ and $Re = 55.0$), the spatial patterns of the cumulative calcite dissolution are similar to those at $Re = 1.0$. However, at $Re = 55.0$, the spatial variation and the difference between the pervious and the impervious case are minor.

3. Fracture with multiple sine wave roughness

For a more complex roughness profile with a combination of multiple sine waves at low Reynolds numbers, $Re = 0.1$, 1.0, and

10.0 (Fig. 11), the streamlines behave similar to the case of the single sine wave geometry. When the fracture walls are impervious, the streamlines Ψ are parallel to the wall geometry as the viscous effects dominate inertial effects [Fig. 11(a)], and the velocity field u , v is periodic along the fracture. In comparison with pervious walls [Fig. 11(b)], the streamlines are not completely parallel to the fracture surface. Ones that start on the fracture surface will compress the streamlines away from the surface. Mean values of u , v increase along the fracture with porous walls as a result of mass conservation. Here, again because of similar structure of results, only the flow field for $Re = 0.1$ is presented.

At high Reynolds numbers, vorticities form in both the pervious and impervious cases as a result of strong inertial effects. Large vorticities are co-located with the long sine wave roughness [Figs. 11(c) and 11(d)].

Similar to the single sine wave roughness, the calcium Ca^{2+} concentration shows a wider boundary layer along the fracture wall in the pervious case compared to the impervious case. This is a result of negative (downward) transverse velocity v in the near wall regions. Figure 12 shows that with solid walls, the transport boundary layer is relatively wide at a low Reynolds number, $Re = 0.1$, and becomes narrower as the Reynolds number increases. In comparison, the influx through the walls helps keep the solute transport boundary layer wide at $Re = 0.1$, 1.0, and 10.0. At $Re = 55.0$, the influx through the walls maintains an observable boundary layer along the fracture despite the local reverse flow, but the width is significantly reduced as a result of large longitudinal velocity [Fig. 12(h)].

Figure 13 compares the point-wise normalized cumulative calcite dissolution for the multi-sine wave profile at different Reynolds numbers. The point-wise mineral dissolution of fracture walls in the case of pervious walls is $\sim 1.2\%$ higher than the case of pervious walls at $Re = 0.1$, 2% higher at $Re = 1.0$, 0.3% higher at $Re = 10.0$, and almost the same at $Re = 55.0$.

The cumulative calcite dissolution in the impervious fracture at a low Reynolds number, $Re = 0.1$, shows local maxima or reaction hotspots at points corresponding to the local minimum cross section of the fracture [Fig. 13(a)]. The cumulative dissolution curve at low Reynolds numbers mirrors the roughness profile. At higher Reynolds numbers, $Re = 1.0$ and $Re = 10.0$, the reaction hotspots become slightly shifted from the local minimum of the fracture aperture [Figs. 13(b) and 13(c)], still representing the roughness profile. This shift means that the dissolution hotspots occur slightly upstream of the minimum fracture aperture, and as a consequence in long time, dissolution will be less effective in smoothing the fracture roughness. It is likely that this pattern leads to an increase in fracture roughness. For the highest Reynolds number, $Re = 55.0$, the cumulative dissolution profile shows no correlation to the roughness geometry [Fig. 13(d)]. This is mainly because of the formation of vortices that affects the dissolution regime on the fracture surface. Moreover, when the fracture walls are permeable, the cumulative calcite dissolution is slightly larger than the corresponding impervious case.

B. pH and effective reaction rate

In the investigated problem, pH increases from 4.2 at the fracture inlet to roughly 4.6 close to the fracture surface. This is almost the same for all the roughness profiles at all Reynolds numbers. The

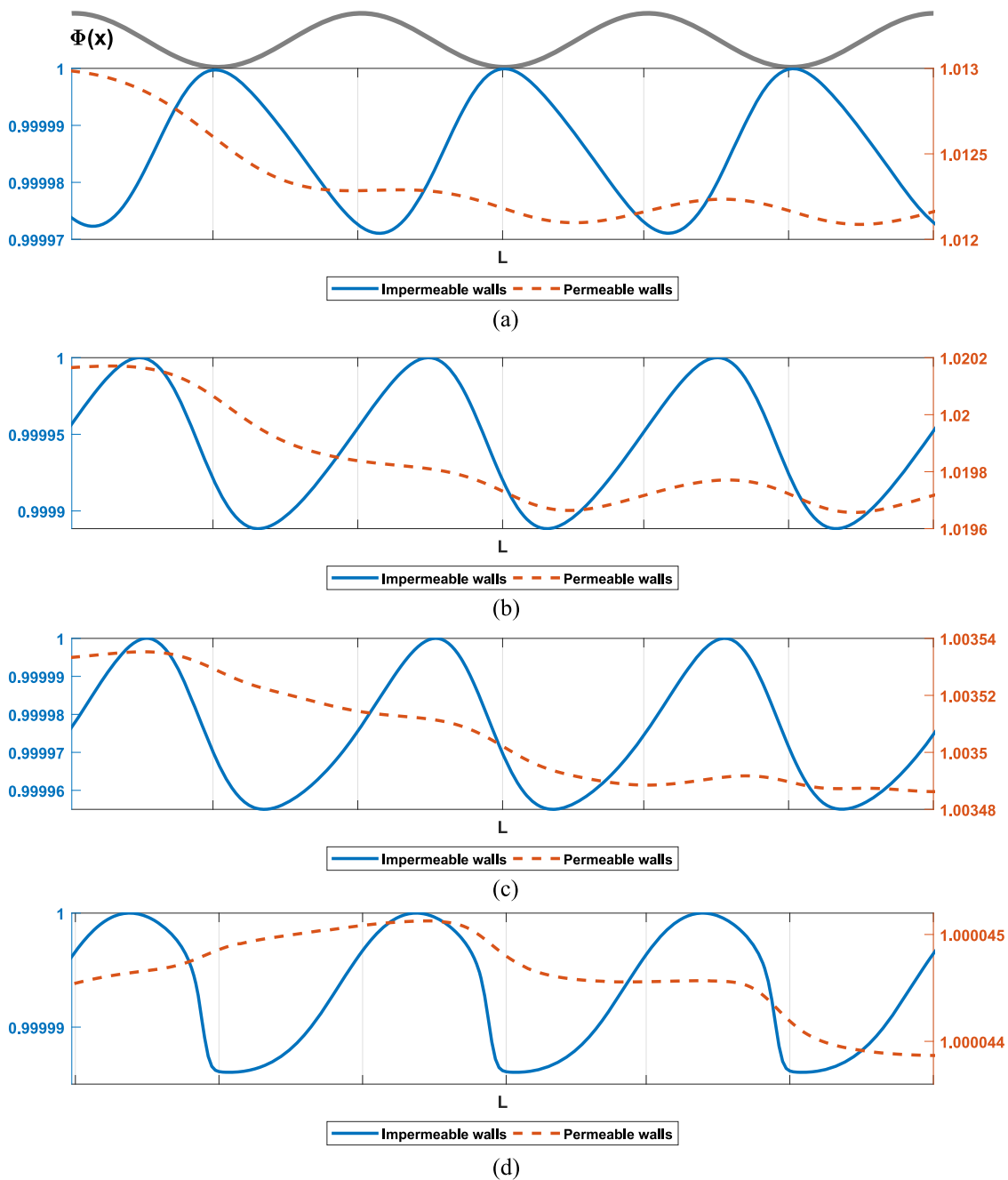


FIG. 10. Cumulative calcite dissolution on the impervious and pervious fracture walls of the single sine wave roughness profile normalized with respect to the maximum of the corresponding impervious fracture wall simulation, $Re = 0.1$ (a), $Re = 1.0$ (b), $Re = 10.0$ (c), and $Re = 55.0$ (d). The blue solid line reads from the left axis and the dashed orange line reads from the right axis. The thick solid gray line represents the upper fracture surface border.

permeable walls show a slightly lower pH because of the more acidic influx through the walls, but the average pH remains close to 4.6.

Effective reaction rates for calcite dissolution from all simulations are presented in Fig. 14. We observe a higher effective reaction

rate in the more rough fractures. Note that $SRF_{\text{flat}} < SRF_{\text{multiple sine}} < SRF_{\text{single sine}}$, according to the values given in Eq. (15). This result is consistent with the effective reaction rates reported in Ref. 52. Both this study and Ref. 52 report the largest increase at the Reynolds

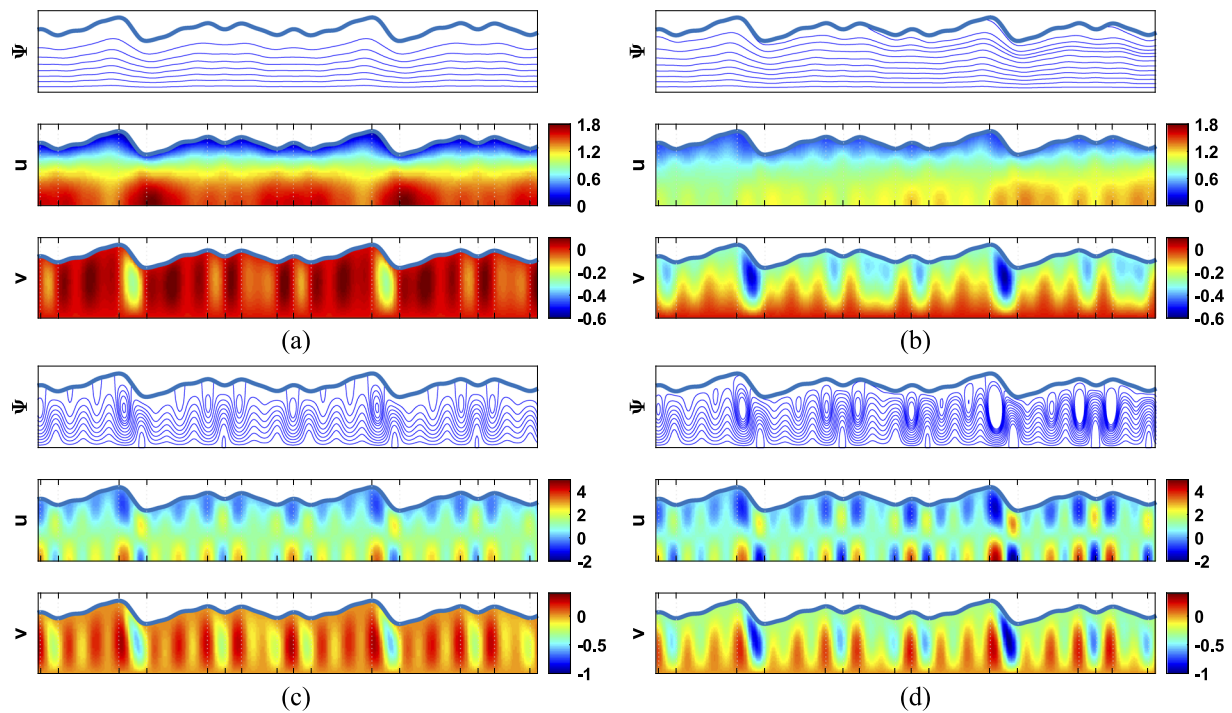


FIG. 11. Flow field within half of a fracture with a multiple sine wave wall geometry at low Reynolds numbers, $Re = 0.1$: impervious (a)/pervious (b) walls, and high Reynolds number, $Re = 55.0$: impervious (c)/pervious (d) walls. The thick solid blue line represents the upper fracture surface border.

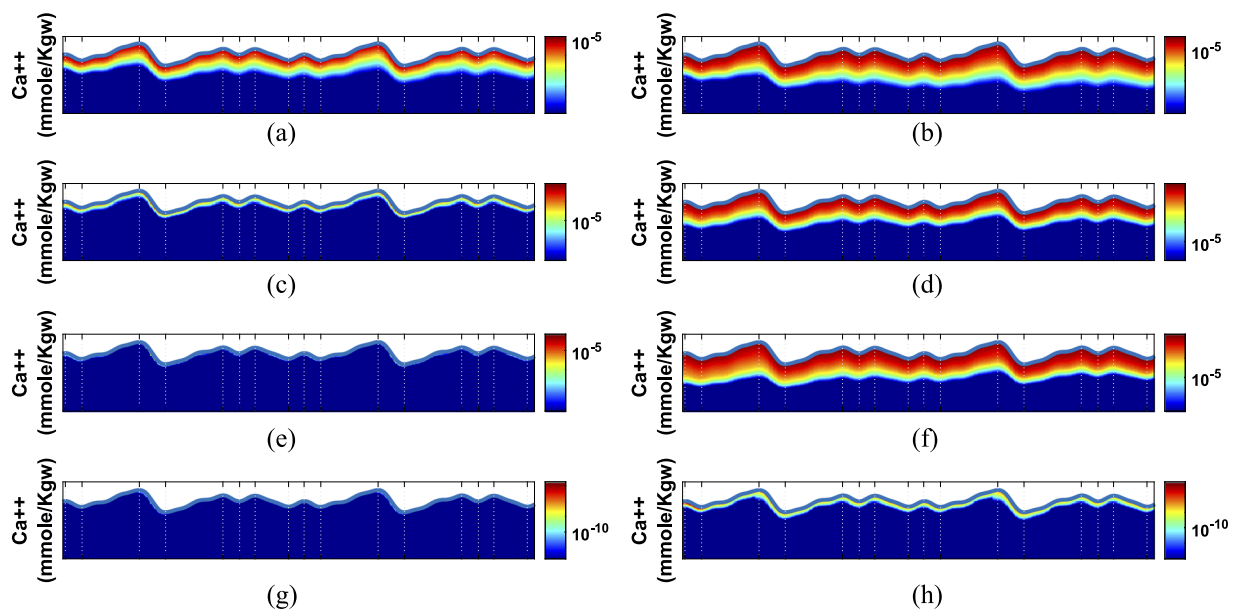


FIG. 12. Calcium ion concentration within half of a fracture with overlaying multiple sine wave roughness geometry at different fluid velocities, $Re = 0.1$: impervious (a)/pervious (b) walls, $Re = 1.0$: impervious (c)/pervious (d) walls, $Re = 10.0$: impervious (e)/pervious (f) walls, $Re = 55.0$: impervious (g)/pervious (h) walls. The thick solid blue line represents the upper fracture surface border.

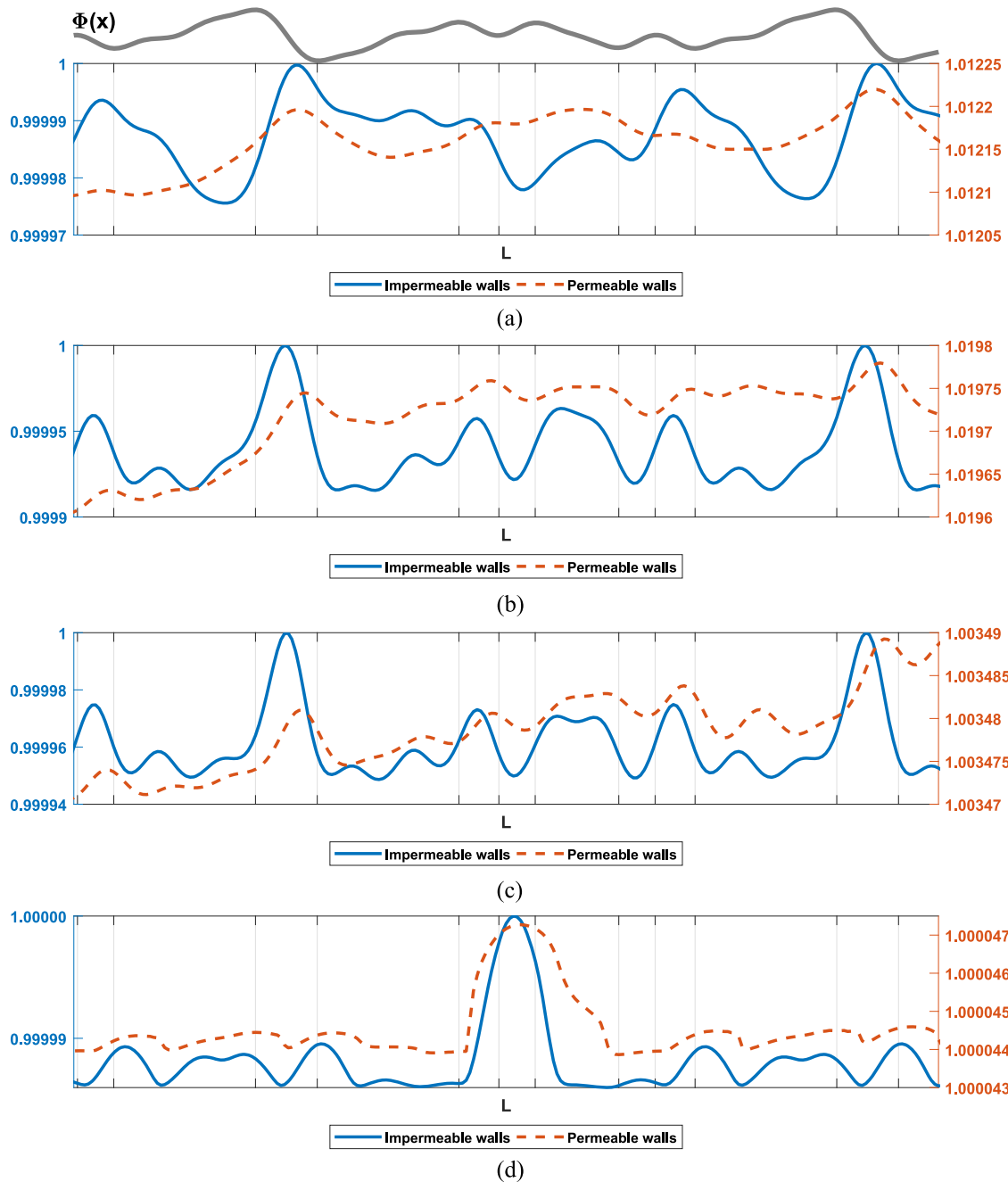


FIG. 13. Cumulative calcite dissolution on the impervious and pervious fracture walls of the multiple sine wave roughness profile normalized with respect to the maximum of the corresponding impervious fracture wall simulation, $Re = 0.1$ (a), $Re = 1.0$ (b), $Re = 10.0$ (c), and $Re = 55.0$ (d). The blue solid line reads from the left axis and the dashed orange line reads from the right axis.

number near $Re = 1.0$. In this work, it was noted that at low Reynolds numbers, $0.01 < Re < 1$, the influx through the fracture walls can improve the reaction rate significantly. At relatively larger Reynolds numbers, the impact of the interporosity fluid exchange through the

walls on the effective reaction rate is less important, but for some roughness profiles still not negligible. These observations are consistent with the point-wise cumulative calcite dissolution as well. This can be related to the formation of vorticities in rough fractures.

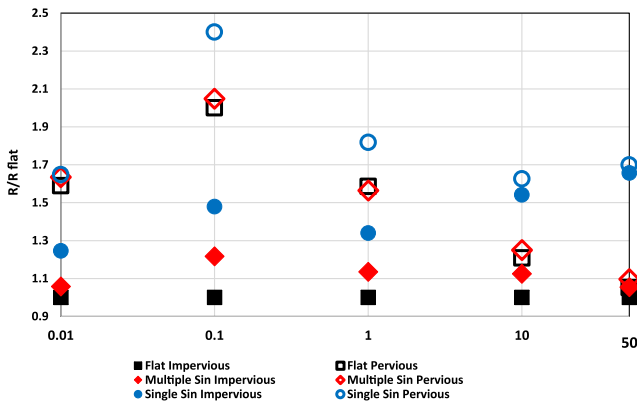


FIG. 14. Effective reaction rate of different roughness profiles at various Reynolds numbers normalized to the effective reaction rate at the corresponding Reynolds number and flat profile.

IV. CONCLUSION

In this study, we developed a new pore-scale modeling framework that allows consideration of reactive transport processes in rough fractures with permeable walls. This capability allows us to simulate systems with more realistic features. Through a test problem of calcite dissolution in CO_2 -acidified fluid, we examined the impacts of fracture roughness, flow velocity, and interporosity fluid exchange on chemical reactions in single fractures. Our simulation results showed that with the interporosity fluid exchange, calcite dissolution is enhanced because an additional influx with the same fluid chemistry as the influent enters into the fracture and changes the local chemical gradients close to the solid–fluid interface. This effect is especially evident at a Reynolds number of 0.1, for which the

effective rate is higher by a factor of 2 in the permeable case than the corresponding impervious case. At high Reynolds numbers, the effect of the interporosity fluid exchange is diminished. Roughness enhances the overall reaction rate at all flow velocities. The spatial distribution of reaction hotspots is also dependent on the interporosity fluid exchange and roughness. For the impervious fractures and at low Reynolds numbers, the local reaction rate is inversely correlated with the fracture aperture, and thus, flow structures are expected to dissolve faster. As a result, the permeability may increase. As the flow velocity increases, the reaction hotspots are shifted from the small aperture locations, which may then lead to an increase in roughness. Qualitatively, similar results were reported by Deng *et al.*⁵² At low Reynolds numbers, they showed higher values for saturation index at the domes of a fracture with impervious walls compared to other locations on the wall, which shows lower dissolution at wider fracture cross sections.

For fractures with pervious walls, the spatial patterns of reaction hotspots are the result of a combination of fracture geometry and influx through the walls. In most cases, an overall trend is observed along the flow direction, implying the dominance of the interporosity fluid exchange, rather than following the fracture geometry.

While the simulations investigated in this study are based on a simple geochemical system and consider a specific interporosity fluid exchange scenario, the model can be extended to other systems. Furthermore, the simulation results highlighted the important role of fracture roughness and interporosity fluid exchange in mineral dissolution and, thus, fracture evolution. Such a modeling capability will help improve our understanding of fractured rock alteration and, thus, develop insights for systems such as the dissolution of limestone rock beneath the ground surface in karst formations by acidified rain water during formation and evolution of sinkholes or the rock mineral dissolution in geologic carbon storage.

APPENDIX: ASYMPTOTIC SOLUTION FOR FLOW FIELD

$$\begin{aligned}
 u(x, \zeta) = & -\frac{3}{2} \frac{F_0}{\phi} (\zeta^2 - 1) + \frac{\omega}{\varepsilon} \left\{ -\frac{3\text{Re} F_0^2 \phi'_y}{280\phi} (7\zeta^6 - 35\zeta^4 + 33\zeta^2 - 5) + \varepsilon \left[\frac{\text{Re} F_0 v^*}{280} (7\zeta^6 - 9\zeta^2 + 2) + \frac{3u^* \phi(0)}{2\phi} (\zeta^2 - 1) + u^* \right] + \varepsilon^2 \dots \right\} \\
 & + \frac{\omega^2}{\varepsilon^2} \left\{ \frac{1}{20} \left(-3F_0 \phi''_{yy} + \frac{9F_0 \phi_y'^2}{\phi} \right) (5\zeta^4 - 6\zeta^2 + 1) + \frac{\text{Re}^2 F_0^3 \phi''_{yy}}{1120} \left(\zeta^{10} - 9\zeta^8 + \frac{138}{5} \zeta^6 - 42\zeta^4 + \frac{9837}{385} \zeta^2 - \frac{1213}{385} \right) \right. \\
 & - \frac{\text{Re}^2 F_0^3 \phi_y'^2}{280\phi} \left(\frac{7}{10} \zeta^{10} - \frac{27}{4} \zeta^8 + \frac{102}{5} \zeta^6 - \frac{57}{2} \zeta^4 + \frac{12333}{770} \zeta^2 - \frac{575}{313} \right) + \varepsilon \left[-\frac{1}{4} (v^* \phi'_y) (5\zeta^4 - 6\zeta^2 + 1) + \frac{\text{Re} u^* \phi'_y F_0}{280} (105\zeta^4 - 126\zeta^2 + 21) \right. \\
 & + \frac{3\text{Re} u^* \phi(0) \phi'_y F_0}{140\phi} (7\zeta^6 - 35\zeta^4 + 33\zeta^2 - 5) + \frac{\text{Re}^2 v^* F_0^2 \phi'_y}{5600} \left(6\zeta^{10} - \frac{75}{4} \zeta^8 - 21\zeta^6 - 1725\zeta^4 + \frac{2759}{11} \zeta^2 - \frac{1961}{44} \right) \left. \right] \\
 & + \varepsilon^2 \left[-\frac{\text{Re} u^* v^* \phi(0)}{280} (7\zeta^6 - 9\zeta^2 + 2) - \frac{\text{Re} u^* v^* \phi}{40} (5\zeta^4 - 6\zeta^2 + 1) + \frac{\text{Re}^2 v^{*2} F_0 \phi}{2800} \left(\frac{\zeta^{10}}{3} - \frac{15}{2} \zeta^8 + 3\zeta^6 + \frac{438}{77} \zeta^2 - \frac{703}{462} \right) \right] + \varepsilon^3 \dots \left. \right\} \\
 & + O\left(\frac{\omega^3}{\varepsilon^3} + \omega\varepsilon\right), \tag{A1}
 \end{aligned}$$

$$v(x, \zeta) = \frac{\omega}{\varepsilon} \left\{ -\frac{3F_0\phi'_y}{2\phi} (\zeta^3 - \zeta) + \varepsilon \frac{3v^*}{2} \left(\frac{\zeta^3}{3} - \zeta \right) + \varepsilon^2 \dots \right\} + \frac{\omega^2}{\varepsilon^2} \left\{ -\frac{3\text{Re } \phi_y'^2 F_0^2}{280\phi} (7\zeta^7 - 35\zeta^5 + 33\zeta^3 - 5\zeta) + \frac{3\text{Re } \phi_{yy}'' F_0^2}{280} (\zeta^7 - 7\zeta^5 + 11\zeta^3 - 5\zeta) \right. \\ \left. + \varepsilon \left[\frac{3u^* \phi'_y \phi(0)}{2\phi} (\zeta^3 - \zeta) + \frac{3\text{Re } v^* \phi'_y F_0}{140} (2\zeta^7 - 7\zeta^5 + 10\zeta^3 - 5\zeta) \right] + \varepsilon^2 \left[\frac{\text{Re } \phi v^{*2}}{280} (-\zeta^7 + 3\zeta^3 - 2\zeta) \right] + \varepsilon^3 \dots \right\} + O\left(\frac{\omega^3}{\varepsilon^3} + \omega\varepsilon\right), \quad (\text{A2})$$

$$\frac{\partial p}{\partial x}(x, \zeta) = -\frac{3F_0}{\phi^3} + \frac{\omega}{\varepsilon} \left\{ \frac{54\text{Re } \phi_y' F_0^2}{35\phi^3} + \varepsilon \left[\frac{3u^* \phi(0)}{\phi^3} - \frac{81\text{Re } v^* F_0}{35\phi^2} \right] + \varepsilon^2 \dots \right\} + \frac{\omega^2}{\varepsilon^2} \left\{ \left(\frac{9\phi_y'^2 F_0}{\phi^3} - \frac{9\phi_{yy}'' F_0}{2\phi^2} \right) \zeta^2 + \frac{3\phi_{yy}'' F_0}{10\phi^2} - \frac{12\phi_y'^2 F_0}{5\phi^3} \right. \\ \left. + \frac{156\text{Re } v^2 F_0^3}{13475} \left(\frac{4\phi_y'^2}{\phi^3} - \frac{3\phi_{yy}''}{\phi^2} \right) + \varepsilon \left[-\frac{6v^* \phi'}{\phi^2} \zeta^2 + \frac{3\text{Re } u^* \phi'_y F_0}{5\phi^2} - \frac{271\text{Re } v^* \phi'_y F_0^2}{1925\phi^2} - \frac{108\text{Re } u^* \phi(0) \phi'_y F_0}{35\phi^3} \right] \right. \\ \left. + \varepsilon^2 \left[\frac{81\text{Re } u^* v^* \phi(0)}{35\phi^2} - \frac{6\text{Re } u^* v^*}{5\phi} - \frac{234\text{Re } v^{*2} F_0}{13475\phi} \right] + \varepsilon^3 \dots \right\} + O\left(\frac{\omega^3}{\varepsilon^3} + \omega\varepsilon\right), \quad (\text{A3})$$

$$\psi = -\frac{3}{2} F_0 \left(\frac{\zeta^3}{3} - \zeta \right) + \omega \left\{ -\frac{3}{280} \text{Re } F_0^2 \phi' (\zeta^7 - 7\zeta^5 + 11\zeta^3 - 5\zeta) \right\} + \omega^2 \left\{ \left(-3F_0 \phi \phi'' + 9F_0 \phi'^2 \right) \left(\frac{\zeta^5}{20} - \frac{\zeta^3}{10} + \frac{\zeta}{20} \right) + \frac{1}{280} \text{Re } F_0^3 \phi \phi'' \right. \\ \left. \times \left(\frac{\zeta^{11}}{44} - \frac{\zeta^9}{4} + \frac{69}{70} \zeta^7 - \frac{21}{10} \zeta^5 + \frac{3279}{154} \zeta^3 - \frac{1213}{154} \zeta \right) - \frac{1}{280} \text{Re } F_0^3 \phi'^2 \left(\frac{7\zeta^{11}}{110} - \frac{3}{4} \zeta^9 + \frac{204}{70} \zeta^7 - \frac{57}{10} \zeta^5 + \frac{4111}{770} \zeta^3 - \frac{575}{313} \zeta \right) \right\} + C + O(\omega^3), \quad (\text{A4})$$

where C is an arbitrary constant.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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